

FlagAI powered by The Call AI

An AI-assisted penalty ruling and rule-citation engine for American football



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Problem Statement

The problem

Football penalty decisions are made in seconds under the pressure of a live game, by officials who must recall hundreds of interlocking rules, exceptions, and enforcement procedures from memory. Even at the NFL level, rule misapplication and inconsistent enforcement continue to change outcomes. At the youth, high school, and small-college levels, where officials are part-time volunteers with far less training, the gap between what the rulebook says and what gets called is wider still.

Why it matters

- Rule misapplication, not judgment on contact, is the category of error the league most often corrects after the fact; one wrong enforcement can decide a game, a playoff berth, or a season.
- Coaches, players, broadcasters, and fans have no fast, trustworthy way to check a call against the actual rule text and its enforcement.
- Officiating associations face shortages and high turnover, so new officials need faster, cheaper ways to learn and verify rule interpretations.

Who is impacted

Game officials and officiating associations at every level; coaches and team analysts preparing challenges; sports media explaining calls to audiences; and the roughly 1.4 million youth and high school players whose games are worked by the least-resourced officials.



Innovation Claim

FlagAI, powered by The Call AI, turns a plain-English description of a football play into a penalty ruling in seconds, with the exact rule, section, and article cited and the enforcement (yardage, spot, and down) computed automatically. Unlike existing AI officiating tools that require stadium camera arrays, it runs a deterministic rule engine built from the official rulebook, using a large language model only to understand the play and explain the answer, so every ruling is fast, consistent, and traceable to the rule text.

What makes it innovative

- **Faster:** rulings in under five seconds instead of minutes of manual rulebook lookup.
- **More consistent:** the same facts always produce the same ruling, because the decision is made by rule code, not by a probabilistic model.
- **Cheaper and more accessible:** no cameras, sensors, or league licensing; works on any browser for a high school crew as easily as a broadcast booth.
- **Explainable:** every output shows the rule text it relied on and what facts would change the call.



Technical Expertise & Skills

Technical domain

Applied AI and software engineering: LLM-based information extraction, rule-based expert systems, and full-stack web development with a secure REST API.

UAT courses that support the MVP

- **Completed / in progress:** CSC382 Data Structures & Algorithms (C++), CSC318 Software Engineering (requirements, UML, Agile, CI/CD), CSC442 Secure Coding, ENG102/ENG330 (technical and research writing), COM226 (presentation and pitch skills), SIP311 Student Innovation Project I.
- **Still needed:** AI-major courses in machine learning and natural language processing (LLM prompting, evaluation, and structured output), a database design course, a web/UX development course, and the remaining SIP sequence through SIP409 for the showcase build.

Technical skills required for the MVP

- **Have now:** Python, C++, C#, FastAPI and REST API design, Git/GitHub, JSON, unit testing, Claude API and agent tooling, Splunk-style log analysis for monitoring, DaVinci Resolve for demo video.
- **Will gain in SIP:** prompt engineering with schema-validated (JSON) LLM output; rule-engine design (encoding rulebook articles as testable condition/penalty records); LLM evaluation and regression testing against a labeled play corpus; SQL database modeling (SQLite → PostgreSQL); React front-end basics; cloud deployment (Docker, Render/Railway) with secure secrets handling; Figma for UI mockups.



Project Description & Scope

FlagAI is a web application and API that accepts a written description of a football play, extracts the relevant facts with an LLM, runs those facts through a deterministic rule engine encoded from the 2026 NFL Rulebook, and returns the ruling, the verbatim rule citation, the enforcement, and a plain-language explanation. Solo project: I design and build every component.

Scope (Iron Triangle)

- **In scope for the MVP:** the 25 most frequently called fouls (holding, false start, offside, pass interference, illegal contact, roughing the passer, etc.) with full enforcement logic; a labeled corpus of 200+ plays for regression testing; a browser UI; a documented REST API; a GitHub repo with CI.
- **Out of scope (future work):** video or computer-vision analysis, live-game integration, NCAA/NFHS rule sets, and mobile apps. Based on peer feedback, the scope was narrowed to the text-in, ruling-out engine so the prototype demonstrably works.

Budget (estimate)

- Claude API usage ~\$20-40/month; hosting (Render/Railway + PostgreSQL) ~\$0-25/month; domain thecallai.com already owned; Figma/GitHub free tiers. Total cash cost through SIP409: under \$600. Labor: 8-10 hours/week.

Timeline

- **SIP311 (Fall 2026):** pitch approval, rule-knowledge-base schema, 10 fouls encoded, parser prompt v1, CLI proof of concept.
- **Next SIP term:** 25 fouls with enforcement, FastAPI backend, 200-play test corpus at ≥90% agreement, React UI.
- **SIP409 (Showcase term):** advisory/explanation layer, user testing with 3-5 officials and coaches, deployment, demo video, and final showcase.



Target Market

Value created

FlagAI gives anyone who needs to understand a penalty a fast, consistent, rule-cited answer, replacing minutes of rulebook searching (or guesswork) with a verifiable ruling in seconds.

Primary users: football officials and officiating associations

- Roughly 2,000–3,000 high school and youth football officials per large state, plus college conference crews; predominantly male, ages 25–65 (Gen X and Millennials), part-time and paid per game, motivated by accuracy and advancement to higher levels. Behaviorally they already use quiz apps and study groups; geographically concentrated in Texas, Ohio, Florida, California, and the Southeast where high school football participation is highest.

Secondary users

- **Coaches and team analysts** (high school through NFL) validating challenges and teaching rules to players.
- **Sports media and content creators** who need to explain a call on air or on social media within minutes.
- **Fantasy and fan communities** (Millennial and Gen Z, mobile-first) who argue calls in real time.

Go-to-market

- Freemium web app for fans; per-seat subscription for officiating associations and coaching staffs; API licensing for media and analytics companies.



Prior Art (1 of 3)

Sony Hawk-Eye Virtual Measurement (NFL, 2025)

- **What it is:** A six-camera 8K optical tracking system adopted league-wide to measure the line to gain in about 30 seconds, replacing the chain crew, with virtual recreations for broadcast.
- **How FlagAI differs:** Hawk-Eye measures physical position with stadium hardware and does not interpret fouls or cite rules. FlagAI needs no cameras: it reasons over a described play and returns the applicable rule and enforcement, so it works at any level of football.

Sony Electronics. (2025, April 1). Sony's Hawk-Eye Innovations selected by the NFL to revolutionize line to gain measurements [Press release]. https://pro.sony/ue_US/press/sonys-hawk-eye-innovations-selected-by-the-nfl-to-revolutionize-line-to-gain-measurements-pr

NFL Next Gen Stats (AWS)

- **What it is:** An RFID player-tracking platform running roughly 75 machine-learning models on AWS to produce real-time analytics such as completion probability and tackle likelihood for broadcasters and teams.
- **How FlagAI differs:** Next Gen Stats predicts performance outcomes from tracking data and is available only to the league and partners. FlagAI is a rules-reasoning tool, produces deterministic rulings rather than probabilities, and is accessible to individual officials and coaches.

O'Neill, S. (2026, February 2). A decade of NFL Next Gen Stats innovation. Amazon Science. <https://www.amazon.science/blog/a-decade-of-nfl-next-gen-stats-innovation>



Prior Art (2 of 3)

CFB Referee (NCAA rules study platform)

- **What it is:** A web-based NCAA college football rules study tool and interactive rulebook with quizzes, approved rulings, and a penalty reference for officials in training.
- **How FlagAI differs:** CFB Referee is a lookup and quiz reference: the user must already know which rule to search. FlagAI starts from the play itself, identifies the foul, and computes the enforcement automatically, then cites the rule.

CFB Referee. (n.d.). CFB Referee: NCAA football officiating platform. Retrieved September 11, 2026, from <https://www.cfbreferee.com/>

Referee Quiz: Football Rules (DartBase AI)

- **What it is:** A mobile trivia app with hundreds of multiple-choice questions on football rules, offering Time Attack, Career, and Sudden Death modes for officials and fans studying the laws of the game.
- **How FlagAI differs:** The app tests memorized knowledge with pre-written questions. FlagAI handles novel, free-text plays and returns a cited ruling instead of a score, making it a decision tool rather than a study game.

DartBase AI. (2026). Referee Quiz: Football Rules (Version updated July 28, 2026) [Mobile app]. Google Play. https://play.google.com/store/apps/details?id=com.refai.ref_ai_app



Prior Art (3 of 3)

Official 2026 NFL Rulebook (NFL Football Operations)

- **What it is:** The league's authoritative online rulebook with field specifications, definitions, all rules and penalty enforcement procedures, and officiating mechanics, published as static searchable text.
- **How FlagAI differs:** The rulebook is the source of truth but offers no reasoning: a reader must find, interpret, and combine multiple articles for a single play. FlagAI encodes those articles as executable rule records so the interpretation and enforcement happen automatically, while still quoting the original text for transparency.

National Football League. (2026). 2026 NFL rulebook. NFL Football Operations. <https://operations.nfl.com/the-rules/nfl-rulebook/>

AI officiating research and commentary (frame-by-frame replay, violation detection)

- **What it is:** Industry analyses describe emerging AI officiating tools built on multi-angle replay analysis and real-time violation detection using computer vision and player tracking.
- **How FlagAI differs:** These systems depend on video and sensor feeds and are aimed at professional leagues. FlagAI deliberately separates rule reasoning from perception: it is the rules brain that any such vision system, or a human official, can query with facts, and it ships without any camera dependency.

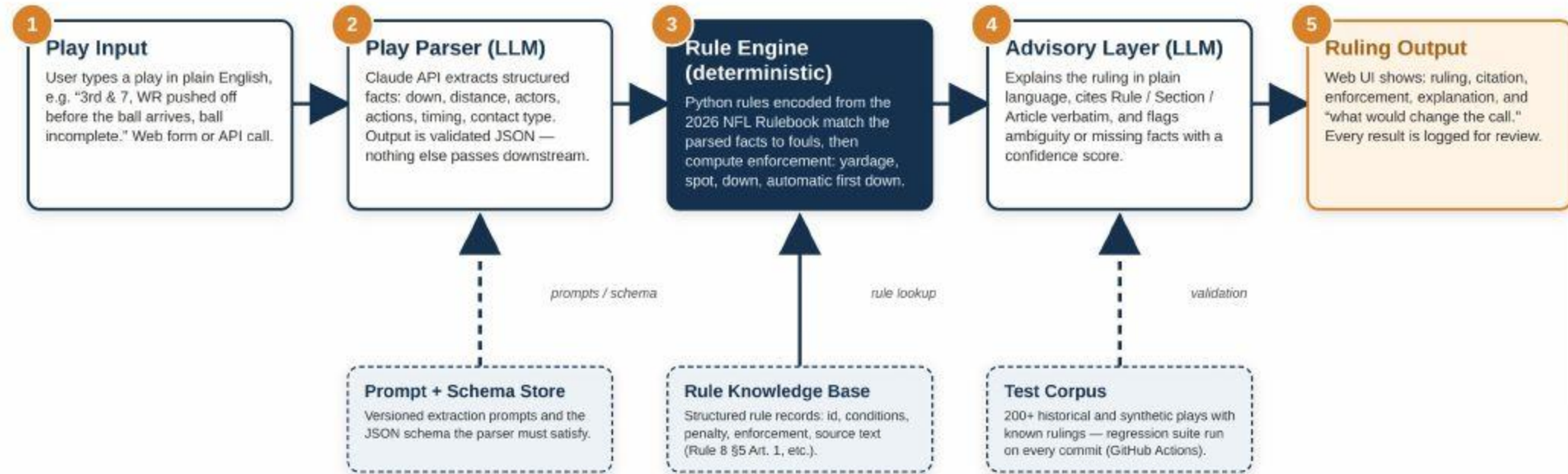
Rodriguez, A. (2025, August 21). Top 5 AI tools for fair play and officiating in sports. Playbook Sports. <https://www.callplaybook.com/reports/top-5-ai-tools-for-fair-play-and-officiating-in-sports>



Proof of Concept: MVP Architecture

FlagAI (The Call AI) — MVP Architecture

Text play description in → deterministic ruling + rule citation out. The LLM never decides the call; it parses and explains.



Stack: Python 3 · FastAPI · Claude API · SQLite/PostgreSQL · pytest · GitHub Actions · React (thin UI) ■ deterministic core □ LLM-assisted ■ user-facing

System flow for the MVP. The LLM parses the play and explains the result; the ruling itself comes from deterministic rule code built from the 2026 NFL Rulebook and validated against a 200-play test corpus.

Early-Stage Prototype: Ruling Interface

FlagAI

The Call AI · rule-engine MVP · v0.1 prototype

Rulebook: 2026 NFL · Engine build 0.1.3

DESCRIBE THE PLAY

3rd and 7 from the offense's 42. QB throws deep down the left sideline. Before the ball arrives, the cornerback grabs the receiver's shoulder and turns him, and the pass falls incomplete. No flag from the crowd's view.

Down: 3rdDistance: 7Spot: OWN 42Qtr: 2

MAKE THE CALL

RECENT PLAYS

#0417 · Illegal formation (6 on line) · Rule 7 §5 Art. 1 · 5 yds

#0416 · No foul · legal contact within 5 yds · Rule 8 §4 Art. 2

#0415 · Roughing the passer · Rule 12 §2 Art. 11 · 15 yds, auto 1st

FLAG · DEFENSE

Defensive Pass Interference

Contact that materially restricts the receiver's opportunity to catch the ball, occurring after the pass and before the ball arrives.

ENFORCEMENT

Spot foul

ball placed at spot of foul

RESULT

Automatic 1st down

1st & 10, OWN 42 — spot

ENGINE CONFIDENCE

92%

RULE CITATION (VERBATIM FROM KNOWLEDGE BASE)

Rule 8, Section 5, Article 1 — It is pass interference by either team when any act by a player more than one yard beyond the line of scrimmage significantly hinders an eligible player's opportunity to catch the ball. Acts include: contact by a defender who is not playing the ball and such contact restricts the receiver's opportunity to make the catch.

WHY THIS CALL · WHAT WOULD CHANGE IT

The parser identified defender-initiated contact (grab + turn) that occurred after the pass was thrown and before arrival, beyond one yard past the line of scrimmage — all three conditions of Rule 8 §5 Art. 1 are met. **The call would change to "no foul"** if the pass were ruled uncatchable, or to **illegal contact (5 yds, auto 1st)** if the contact had occurred before the ball was thrown.

Mockup of the FlagAI web interface: play description in, ruling with rule citation, enforcement, confidence, and "what would change the call" out. Built from the CSC318 SRS use cases; a CLI version of the rule engine is the first SIP311 build target.